Integrated PDE-Agent Workflow for Open Venue Cooling Assessment and Optimization

External Inputs

venue geometry, meteorological data,
cooling device parameters, pedestrian settings

Stage 1: Physics & Pedestrian Simulation

PDE Advection-Diffusion Solver
+ Dual-mode Pedestrian Agent Simulation

Compute perceived temperature θ

(from T, RH, wind speed)

Two-sided discomfort function integration

Calculate Cooling Efficiency

(benefit / energy ratio)

Stage 2: Multi-Criteria Evaluation System

Cooling Efficiency

Average thermal comfort / total discomfort

Pedestrian waiting experience

Total energy consumption

Installation & operation cost

Safety (slip hazard & evacuation)

Iterative optimization loop

Entropy Weight Method + TOPSIS
→ Comprehensive Cooling Effectiveness Score

Stage 3: Surrogate-assisted Layout Optimization

Random Forest Surrogate Model

(to accelerate PDE simulation)

Simulated Annealing Optimizer

Output: Optimal device configuration
& actionable planning recommendations